Roof Gantry System

Senior Design Project Idea by Jael McBride

Not Everyone has a team to help lift

Small businesses and solo hobbyists tend to struggle with moving heavy machinery or work pieces, as not everyone has physical capability to safely move them.



Why this matters

- This problem creates a major barrier for anyone working in hands on or tool intensive environments.
- Whether it's woodworking, large-format CNC machining, automotive repair, or any craft that requires frequent movement of heavy materials, the lack of an efficient and safe way to transport these items slows down workflows, and limits what people can build.

How is this handled currently?





Currently

1. A factory roof gantry system
- too large scale and highly expensive for our needs

2. Smaller tools and lifts that are for specific applications

   (i.e. engine cranes and so on)

- small scale and too specific
- Have to be manually dragged around



Initial direction

The plan is to make a product worth being proud of and usable in a wider range of applications then what's currently on the market.

Examples

- A customizable track and frame to fix various room sizes

- An attachment platform to use other tools for more unique applications

(I.E. camera and light mount & so on)

Why join us?

This year long project is to make something that increases the workflows and limits of smaller creators, all the way from small business owners and shops to passionate hobbyists.

